Practical 6

Task 1

My final year project focuses on algorithmic bias in facial detection systems. Several forms of bias are particularly relevant to this topic, as facial detection technologies are known to perform unevenly across different demographic groups.

One significant risk is sampling and selection bias. Facial detection models are often trained on datasets that over represent certain groups, particularly lighter skin tones, younger individuals and Western populations. If training or evaluation data lacks sufficient diversity, the system may perform well for some groups while consistently underperforming others. This reflects the lecture discussion of WEIRD data and insufficient representation.

Another important concern is historical and systemic bias. Existing social and structural inequalities can be embedded within datasets, even when data collection appears neutral. For example, publicly available image datasets may reflect existing patterns of visibility and exclusion, which can lead to biased model outcomes. As discussed in the lectures, such biases can persist even with technically correct data processing.

Algorithmic bias itself is also relevant. Design choices such as feature selection, model architecture and optimisation metrics can favour majority groups, particularly when accuracy is prioritised over fairness. Aggregation bias may occur if performance metrics are averaged across populations, masking poor performance for specific subgroups.

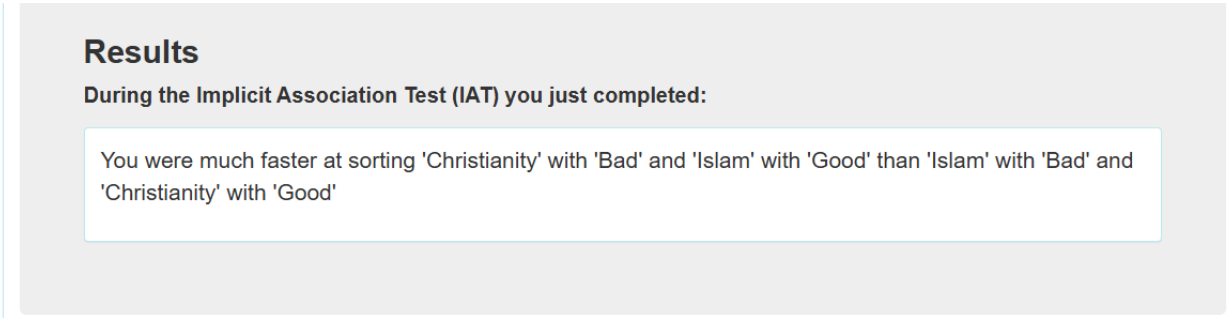
There is also a risk of confirmation bias during evaluation. I may unintentionally focus on results that support the expectation of bias while overlooking cases where the model performs fairly. In addition, automation bias could arise if outputs from established facial detection tools are treated as authoritative rather than critically assessed.

To mitigate these biases, I would use diverse and balanced datasets where possible and evaluate model performance separately across demographic groups rather than relying on aggregate accuracy. Fairness metrics would be incorporated alongside traditional performance measures. Model design choices would be documented transparently, and evaluation criteria would be defined in advance to reduce confirmation bias. Peer review and supervisor feedback would further help challenge assumptions.

Any remaining limitations would be clearly communicated in the project report. This would include acknowledging dataset constraints, potential sources of bias and the limits of fairness evaluation. Clearly communicating these issues is essential to ensure ethical transparency and responsible interpretation of results.

Task 2

Taking the tests



I completed an Implicit Association Test exploring associations between religious groups and evaluative concepts. As a Muslim, the result was unexpected, as it did not align with my conscious beliefs or personal identity. This highlighted one of the key points discussed in the lecture on bias, namely that implicit associations can arise from broader cultural exposure, media narratives, or repeated societal messaging rather than deliberate attitudes or values.

The test also required me to provide contextual information such as age and background, which reinforced how personal and situational factors may influence reaction times. Importantly, the IAT itself acknowledges that results can fluctuate and are affected by variables such as task design, familiarity with the controls, fatigue, and time pressure. This supports existing criticism of the test, which argues that it measures momentary cognitive associations under artificial conditions rather than stable beliefs or real world behaviour.

For these reasons, I do not interpret the result as evidence of personal prejudice. Instead, it serves as an illustration of how unconscious associations can exist even when individuals actively reject biased views. This insight is particularly relevant to my final year project on algorithmic bias in facial detection, where similar unconscious biases can influence data labelling, system design, and evaluation decisions. The exercise reinforced the importance of recognising that both humans and systems are shaped by their environments, and that ethical system design must account for these influences rather than assuming objectivity.